

LAGOS IKEJA, Nigeria

WMO: 652010

Lat: 6.577N

Lon: 3.321E

Elev: 41

StdP: 100.83

Time Zone: 1.00 (E01)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.1	21.9	10.8	8.1	29.7	14.1	10.1	28.4	10.0	27.3	9.3	27.5	1.4	0	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.9	34.8	25.9	34.1	26.1	33.7	26.2	28.3	32.0	27.9	31.7	27.6	31.3	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.1	30.1	27.0	22.8	29.9	26.7	22.4	29.6	91.6	32.2	89.4	32.0	88.0	31.6	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.8	6.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	28.3	29.2	29.6	29.0	28.1	26.9	26.2	26.0	26.4	27.0	28.2	28.5	
	DBStd	1.62	1.37	1.29	1.04	1.39	1.26	1.17	0.90	0.78	0.84	1.09	0.94	1.01	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6486	568	537	606	571	563	506	503	496	491	526	546	574	
	CDD18.3	3444	310	304	348	321	304	256	244	238	241	268	296	316	
	CDH23.3	34167	3236	3466	4169	3786	3297	2281	1847	1613	1812	2341	3033	3288	
	CDH26.7	12228	1366	1479	1864	1658	1236	588	320	248	338	658	1124	1349	
Wind	WSAvg	2.3	1.9	2.4	2.4	2.8	2.2	2.3	2.8	3.1	2.6	2.1	1.8	1.8	
Precipitation	PrecAvg	1569	14	37	79	159	206	318	242	105	178	149	57	18	
	PrecMax	2616	61	188	250	389	450	731	583	468	419	324	137	98	
	PrecMin	872	0	0	6	11	36	118	23	3	28	37	1	0	
	PrecStd	354	14	41	50	78	87	142	165	109	99	64	35	26	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	35.7	35.1	35.0	34.0	32.5	30.9	30.3	31.2	32.5	34.2	35.1	
		MCWB	24.4	25.1	26.7	27.4	27.3	26.7	25.8	24.9	26.0	27.1	25.8	24.8	
	2%	DB	34.2	34.2	34.2	34.1	33.1	31.2	30.0	29.7	30.2	31.8	33.2	34.1	
		MCWB	24.7	26.4	27.1	27.3	27.2	26.5	25.6	24.8	25.6	26.7	26.4	24.8	
	5%	DB	33.5	33.8	33.7	33.2	32.2	30.4	29.2	29.0	29.3	30.9	32.3	33.3	
		MCWB	24.6	26.5	27.0	27.0	26.8	26.2	25.2	24.7	25.3	26.4	26.5	25.1	
	10%	DB	32.6	32.8	33.0	32.8	31.8	29.8	28.3	28.1	28.8	30.0	31.7	32.5	
		MCWB	24.3	26.5	26.8	26.9	26.7	26.0	24.9	24.4	25.2	26.1	26.5	25.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	28.7	29.0	28.7	28.2	27.5	26.5	25.9	26.7	27.7	28.2	28.6
			MCDB	31.7	31.9	32.6	32.7	31.8	30.4	29.2	28.5	29.6	30.9	32.1	31.9
2%		WB	27.5	27.8	28.1	28.0	27.6	26.9	26.0	25.4	26.1	27.1	27.7	27.7	
		MCDB	31.2	31.6	32.1	31.9	31.4	30.0	28.7	28.0	28.9	30.5	31.4	31.4	
5%		WB	26.9	27.4	27.6	27.6	27.2	26.5	25.6	25.0	25.7	26.6	27.3	27.2	
		MCDB	30.4	31.0	31.4	31.3	30.8	29.4	28.1	27.8	28.4	29.8	30.9	30.9	
10%		WB	26.5	27.0	27.2	27.3	26.9	26.1	25.3	24.7	25.4	26.2	26.9	26.7	
		MCDB	29.7	30.4	30.7	30.7	30.3	28.8	27.6	27.2	27.8	29.1	30.3	30.1	
Mean Daily Temperature Range	5% DB	MDBR	10.1	8.3	7.9	7.8	7.4	6.0	5.1	5.1	5.6	6.9	8.2	9.5	
		MCDBR	10.6	8.8	8.3	8.3	8.0	7.2	6.1	6.0	6.5	7.8	8.9	10.4	
		MCWBR	4.2	2.9	3.1	3.1	3.3	3.4	2.6	2.5	2.9	3.7	3.8	3.8	
	5% WB	MCDBR	9.0	8.1	7.8	7.8	7.6	6.6	5.5	5.3	6.0	7.3	8.1	8.7	
		MCWBR	3.9	3.4	3.4	3.3	3.6	3.4	2.6	2.4	2.9	3.8	3.9	3.8	
Clear-Sky Solar Irradiance	taub	0.844	0.852	0.833	0.683	0.572	0.535	0.531	0.538	0.493	0.524	0.622	0.728		
	taud	1.481	1.481	1.499	1.762	1.986	2.101	2.114	2.084	2.237	2.134	1.844	1.654		
	Ebn at Noon	564	578	596	683	744	762	770	779	827	802	711	628		
	Edh at Noon	304	310	307	232	181	158	158	166	144	159	209	251		
All-Sky Solar Radiation	RadAvg	4.92	4.84	5.05	5.11	4.68	3.90	3.83	3.99	4.14	4.63	5.01	4.92		
	RadStd	0.20	0.29	0.26	0.21	0.18	0.30	0.36	0.26	0.25	0.18	0.17	0.17		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.28	+0.31	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page